PRACTICE SET – 2

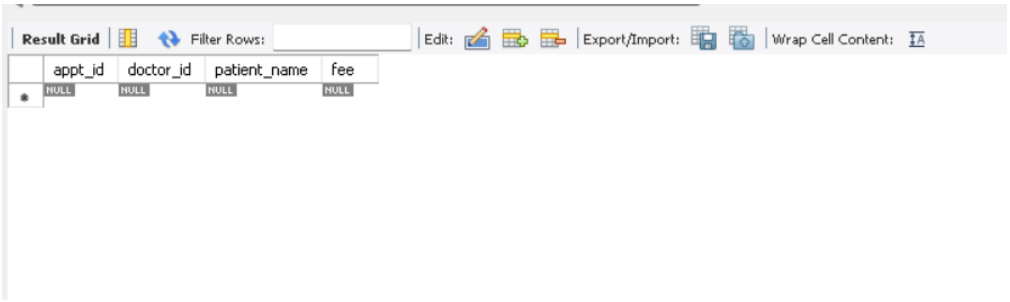
1. **Create the hospital database and both tables exactly as shown.**

```sql
CREATE DATABASE hospital ;
USE hospital ;

CREATE TABLE doctors (
doctor_id INT PRIMARY KEY ,
dname VARCHAR (50) NOT NULL ,
department VARCHAR (30)
);


CREATE TABLE appointments (
appt_id INT PRIMARY KEY ,
doctor_id INT ,
patient_name VARCHAR (50) ,
fee INT ,
FOREIGN KEY ( doctor_id ) REFERENCES doctors ( doctor_id )
ON DELETE CASCADE
);
```

DATABASE OUTPUT:

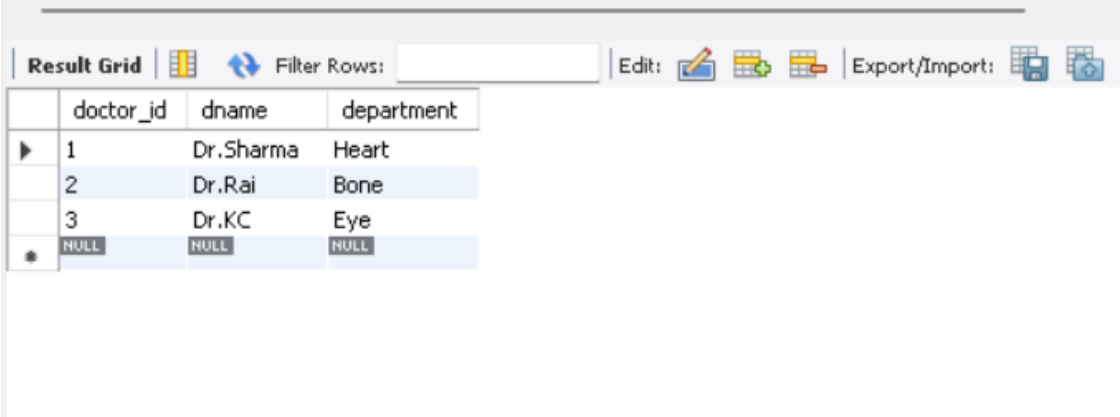




2. **Insert 3 doctors: (1, Dr. Sharma, Heart), (2, Dr. Rai, Bone), (3, Dr. KC, Eye).**

```
23 •   INSERT INTO doctors
24     VALUES
25     (1,"Dr.Sharma","Heart"),
26     (2,"Dr.Rai","Bone"),
27     (3,"Dr.KC","Eye");
28
29 •   SELECT * FROM doctors ;
30
```

DATABASE OUTPUT:



doctor_id	dname	department
1	Dr.Sharma	Heart
2	Dr.Rai	Bone
3	Dr.KC	Eye
NULL	NULL	NULL

3. **Try to insert a doctor with doctor id = 2 again. Write down the error.**

```
30
31 •    INSERT INTO doctors
32      VALUES
33      (2,"Dr.Amatya","Kidney");
34
```

DATABASE OUTPUT:

```
16  11:27:35  INSERT INTO doctors VALUES (2,"Dr.Amatya","Kidney")
```

4. **Try to insert a doctor with NULL id, and one with NULL name. Which constraint blocks each?**

```
34
35 •    INSERT INTO doctors
36      VALUES
37      (NULL, "Dr.Ram","Ear");
38
```

DATABASE OUTPUT:

```
18  11:33:32  INSERT INTO doctors VALUES  (NULL, "Dr.Ram","Ear")
19  11:34:31  INSERT INTO doctors  VALUES (4,NULL,"Throat")
```

5. For each real system, decide a good PK and say why name is not enough: patients in this hospital, products in Daraz , citizens of Nepal.

```
In Hospitals , Patient ID can be the best option to keep as the primary key as every patient id is unique and patients can have same name
too so keeping patient name as primary key cannot be an option

For products in Daraz , Product ID can be the best option to keep as the primary key as products might have similar or same names so keeping
the product name as primary key cannot be an option , every product id is unique and different so it is the best option

For citizens of Nepal , Citizenship Number can be the best option as citizenship number are unique, different and cannot be the same
```

6 .Insert 4 appointments for existing doctors, two of them for doctor 1. Why is the repeated doctor id allowed?

```
42
43 •    INSERT INTO appointments
44      VALUES
45      (100,1,"Michael",800),
46      (101,1,"Jackson",800),
47      (102,2,"Jordan",500),
48      (103,2,"Faraday",500);
49 •    SELECT * FROM appointments ;
50
```

DATABASE OUTPUT:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

appt_id	doctor_id	patient_name	fee
100	1	Michael	800
101	1	Jackson	800
102	2	Jordan	500
103	2	Faraday	500
NULL	NULL	NULL	NULL

7. Try to insert an appointment with doctor id = 99. What happens and why is this good?

```
51 •    INSERT INTO appointments
52      VALUES
53      (104,99,"Mike",850);
54
```

DATABASE OUTPUT:

```
27  14:00:54  INSERT INTO appointments VALUES (104,99,"Mike",850)
```

```
7. It prevents the unnecessary appointment from being created
```

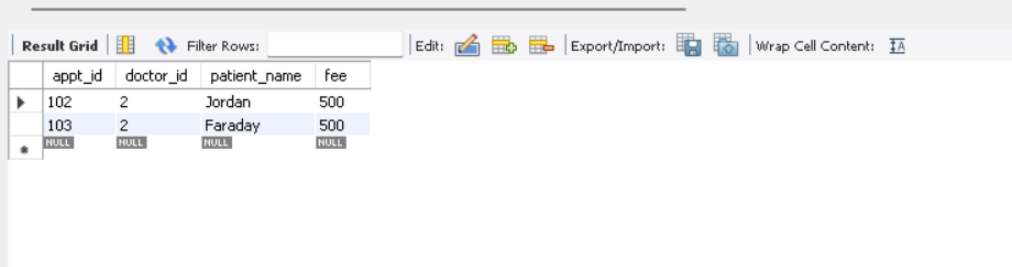
8. Which table is parent and which is child? Where does the FK live?

```
8. doctors is the parent table and appointments is the child table.
```

9. Delete doctor 1. What happened to doctor 1's appointments, and which FK option caused that?

```
55 •    DELETE FROM doctors WHERE doctor_id = 1;
```

DATABASE OUTPUT:



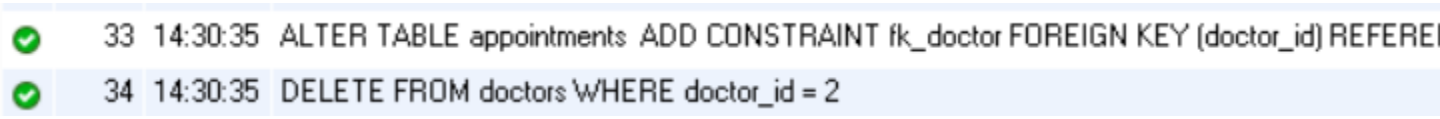
appt_id	doctor_id	patient_name	fee
102	2	Jordan	500
103	2	Faraday	500
NULL	NULL	NULL	NULL

```
9.Doctor 1's appointments are deleted too
```

10. Recreate the FK with RESTRICT instead. Now try deleting a doctor who has appointments. What happens?

```
58
59
60 •  ALTER TABLE appointments
61    ADD CONSTRAINT fk_doctor
62    FOREIGN KEY (doctor_id) REFERENCES doctors ( doctor_id)
63    ON DELETE RESTRICT ;
64
65 •  DELETE FROM doctors WHERE doctor_id = 2;
```

DATABASE OUTPUT:

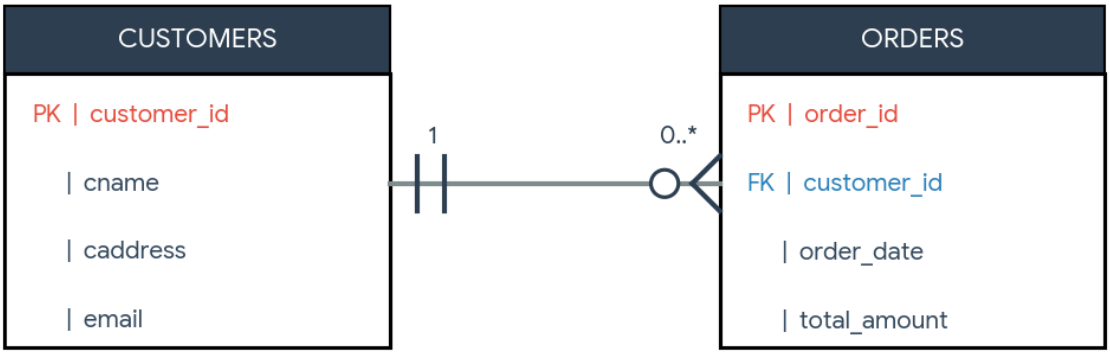


11. Real life: Daraz has customers and orders. Draw the parent, the child, and the FK column.

```
66
67 •  CREATE TABLE customers (
68    customer_id  INT PRIMARY KEY ,
69    cname VARCHAR(30) ,
70    caddress VARCHAR(60),
71    email VARCHAR(70)
72    );
73
74 •  CREATE TABLE orders (
75    order_id INT PRIMARY KEY ,
76    customer_id INT ,
77    order_date VARCHAR(40),
78    total_amount INT ,
79
80    FOREIGN KEY (customer_id) REFERENCES customers ( customer_id)
81    );
82
```

DATABASE OUTPUT:

```
36  14:39:41  CREATE TABLE customers ( customer_id  INT PRIMARY KEY , cname VARCHAR(30) ,  caddre
37  14:39:41  CREATE TABLE orders ( order_id INT PRIMARY KEY ,  customer_id INT ,  order_date VARCHA
```



12. Why can an FK repeat and be NULL while a PK cannot? One sentence each.

```
12. Primary Key cannot be repeated or null as it uniquely identifies e
    Foreign Key can be repeated or null as it has one-to-many relation
link to the parent table is optional
```